

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO5 ASSESSMENT TOOL 1 – Research Paper Review

Course Code:	CSA1007	Course Title:	Software Engineering
CO Assessed:	CO5	Assessment:	Assessment Tool 1 – Research Paper Review
Weightage:	20%	Total Marks:	25
CO5 Statement:	Evaluate emerging technologies and societal impacts, maintaining and evolving software systems responsibly		
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Date:	01-09-2026	Duration:	60 minutes

Code of Conduct

I RISHIK JK (192511052) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: **RISHIK JK**

CO5-AT1-Research Paper Review/ LITERATURE SURVEY

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COURSE CODE: CSA1007

DATE: 01/09/2026

RESEARCH PAPER REVIEW / LITERATURE SURVEY

Project Title: Smart Renewable Battery Recycling Platform

1. Introduction

The increasing use of electric vehicles and electronic devices has resulted in a rapid growth of end-of-life batteries. Improper handling of these batteries can create environmental pollution, safety hazards, and loss of valuable materials such as lithium, cobalt, nickel, and other metals.

Battery recycling therefore requires efficient collection, monitoring, sorting, recycling, tracking, and resource recovery. Recent research has explored technologies such as blockchain, machine learning, online sensing, and intelligent supply-chain management to improve battery recycling.

This literature survey reviews existing research related to smart battery recycling and identifies the technologies, methodologies, findings, limitations, and research gaps that can support the development of the proposed Smart Renewable Battery Recycling Platform.

2. Objectives of the Literature Survey

The main objectives of this literature survey are:

- To study recent research related to battery recycling and management.
- To understand technologies used for battery tracking, recycling, and safety.
- To analyze the use of blockchain and machine learning in battery recycling.
- To compare different existing approaches and their benefits.
- To identify limitations and research gaps in existing systems.
- To determine possible improvements for the proposed platform.

3. Selection of Research Papers

Five research papers were selected based on their relevance to battery recycling, traceability, recycling optimization, battery health analysis, and intelligent automation.

- The selected papers mainly cover:
- Blockchain-based battery traceability.
- Machine learning for battery health diagnosis.
- Automated battery disassembly.
- Machine learning for recycling optimization.
- Closed-loop battery recycling and supply-chain management.

These papers provide different technological approaches that are useful for understanding how a smart battery recycling platform can be developed.

4. Review of Selected Research Papers

4.1 Paper 1 – A Blockchain-Enabled Framework for Reverse Supply Chain Management of Power Batteries

This research proposes a blockchain-based framework for managing the reverse supply chain of used power batteries. The main objective is to improve transparency and traceability throughout the battery recycling process.

The proposed system combines blockchain with Internet of Things technology to track batteries. It uses a four-layer architecture, a double-chain storage mechanism, an information-tracing module, and smart contracts for battery trading.

The study shows that blockchain can improve information sharing between stakeholders and provide trustworthy battery information. The system also helps trace the source and destination of used batteries and improves control over recycling activities.

One limitation is that implementing blockchain and IoT technologies can increase system complexity and infrastructure requirements.

4.2 Paper 2 – Rapid and Sustainable Battery Health Diagnosis for Recycling Pretreatment Using Fast Pulse Test and Random Forest Machine Learning

This research focuses on identifying the State of Health of retired lithium-ion batteries before recycling.

The proposed approach combines a fast pulse test with the Random Forest machine learning algorithm. The study evaluates 442 retired batteries obtained from laboratory testing and actual electric vehicle usage.

The approach helps determine battery condition without requiring large amounts of historical battery data. It can reduce testing time, cost, and environmental impact while helping recyclers decide whether a battery should be reused or recycled.

However, battery characteristics can vary significantly across manufacturers and operating conditions. Therefore, maintaining prediction performance across many different battery types remains challenging.

4.3 Paper 3 – A Novel Disassembly Process of End-of-Life Lithium-Ion Batteries Enhanced by Online Sensing and Machine Learning Techniques

This research focuses on improving the safety and automation of lithium-ion battery disassembly.

The proposed platform uses computer vision to identify different batteries and thermal cameras to monitor temperature during disassembly. Machine learning is used to predict temperature behaviour, while a neural-network model is used to identify cutting defects.

The major benefit of this system is that it reduces direct human exposure to hazardous battery materials. Real-time sensing and closed-loop control also improve the safety and quality of the battery disassembly process.

A major challenge is that automated disassembly systems require specialized sensors, cameras, machinery, and trained machine-learning models, which can increase implementation cost.

5. Paper 4 – Predictive Precision in Battery Recycling: Unveiling Lithium Battery Recycling Potential Through Machine Learning

This research investigates how machine learning can improve lithium battery recycling.

The study focuses on predicting recycling viability, optimizing recycling processes, and improving resource recovery. It presents a structured machine-learning framework that includes data preprocessing, feature engineering, model development, and evaluation.

The study indicates that data-driven techniques can help recycling organizations make better decisions and improve the efficiency of resource recovery.

However, the effectiveness of machine-learning models depends strongly on the quality and availability of battery data. Limited real-world datasets and differences between battery chemistries can affect prediction accuracy.

6. Paper 5 – Blockchain-Enabled Closed-Loop Supply Chains for Sustainable Electric Vehicle Battery Recycling

This research studies the use of blockchain in closed-loop supply chains for electric vehicle battery recycling.

The proposed model considers manufacturers, battery suppliers, recyclers, and blockchain service providers. It uses a Stackelberg game-theoretic approach to analyze different contractual strategies between these stakeholders.

The results show that blockchain-based information sharing can improve traceability, recycling efficiency, stakeholder coordination, and sustainability. Different contractual strategies can also influence battery recovery and supply-chain performance.

The study mainly concentrates on supply-chain decisions and economic relationships. A complete operational software platform combining inventory, hazard prediction, logistics, recycling operations, and sustainability reporting still requires further development.

7. Critical Analysis of Existing Approaches

The reviewed research demonstrates that several emerging technologies can improve battery recycling. Blockchain is effective for maintaining transparent and traceable battery information, while machine learning can support battery health prediction, recycling decisions, and process optimization.

Online sensing, computer vision, and intelligent automation can also improve worker safety by reducing direct interaction with hazardous batteries.

However, most existing approaches concentrate on one particular part of the recycling process. Some systems mainly focus on traceability, while others concentrate on battery health prediction, automated disassembly, or supply-chain optimization.

There is therefore an opportunity to integrate these capabilities into a common software platform that manages the complete battery recycling process.

8. Limitations and Challenges

The major limitations identified from the literature are:

- High cost of implementing advanced sensors and automation equipment.
- Limited availability of standardized real-world battery datasets.
- Differences in battery chemistry, capacity, age, and manufacturer.
- Complexity of implementing blockchain across multiple organizations.
- Privacy and security concerns when sharing battery information.
- Difficulty in integrating different recycling stakeholders into one system.
- Machine-learning models require reliable and sufficient training data.
- Existing approaches often address only individual parts of the recycling lifecycle.

These limitations should be considered while designing a practical battery recycling platform.

09. Identified Research Gap

The literature shows significant progress in battery traceability, machine-learning-based prediction, automated disassembly, and recycling supply-chain optimization.

However, there is still a need for an integrated platform that combines these functions and manages the complete recycling lifecycle.

The identified research gap is the lack of a centralized smart platform that can simultaneously:

- Monitor battery inventory.
- Manage recycling operations.
- Predict hazardous battery conditions.
- Track battery transportation and logistics.
- Maintain traceable recycling information.
- Generate sustainability reports.
- Notify recycling operators about important events.

The proposed Smart Renewable Battery Recycling Platform attempts to address this gap by bringing the major recycling management activities into a single system.

10. Future Research Directions

Future development can improve the proposed system using the following approaches:

10.1 AI-Based Hazard Prediction

Machine-learning models can analyze battery temperature, voltage, condition, and other sensor information to identify hazardous conditions before an accident occurs.

10.2 IoT-Based Real-Time Monitoring

IoT sensors can continuously monitor batteries during storage, transportation, and recycling operations.

10.3 Blockchain-Based Traceability

Blockchain can maintain secure records of the battery lifecycle from manufacturing to recycling and resource recovery.

10.4 Automated Battery Classification

Computer vision and machine-learning models can automatically identify battery types and determine suitable recycling methods.

10.5 Smart Logistics Management

Real-time location tracking and route optimization can improve the transportation of used batteries between collection centers and recycling facilities.

10.6 Sustainability Analytics

Future systems can calculate recycling rates, recovered materials, waste reduction, and environmental impact to support sustainability reporting.

11. Relevance to the Proposed System

The reviewed research provides useful technologies and ideas for the Smart Renewable Battery Recycling Platform.

Blockchain research supports the development of battery traceability and secure lifecycle records. Machine-learning research provides methods for battery condition assessment and hazard prediction. IoT and sensing technologies can support real-time battery monitoring, while intelligent supply-chain approaches can improve recycling logistics.

By combining suitable concepts from these existing studies, the proposed system can provide a more integrated approach to battery recycling management.

12. Conclusion

The literature survey shows that emerging technologies such as machine learning, blockchain, IoT, computer vision, and intelligent automation can significantly improve battery recycling.

Existing studies provide effective solutions for specific areas such as battery health diagnosis, traceability, automated disassembly, and supply-chain management. However, these functions are often developed separately.

The proposed Smart Renewable Battery Recycling Platform aims to provide a centralized solution that integrates battery inventory monitoring, recycling management, hazard prediction, logistics tracking, sustainability reporting, and operator notifications. Such an integrated approach can improve recycling efficiency, operational safety, resource recovery, and environmental sustainability.

13. References

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